

Project Demonstration

Date	11 July 2026
TeamID	XXXXXX
Project Name	EduGenie–Google Gemini Powered Educational Assistant
Maximum Marks	5Marks

Project Demonstration Overview

This section demonstrates the complete workflow of **EduGenie – AI-Powered Educational Assistant**. The application provides intelligent educational support by answering academic questions, explaining complex concepts in simple language, generating topic-based quizzes, summarizing educational content, and creating personalized learning paths. The system integrates **Fast API**, **HTML/CSS** and **Generative AI models** to deliver a responsive and interactive learning experience. The demonstration verifies that all core modules operate successfully and satisfy the project requirements.

1. System Architecture

The architecture of EduGenie consists of the following components:

- Frontend: HTML, CSS, and Jinja 2 templates provide an interactive and user–friendly interface.
- Backend: FastAPI handles HTTP requests, API routing, and communication between modules.
- AI Models: GeminiAPI and LaMini–Flan–T5 process educational queries and generate intelligent responses.
- Utility Modules: Manage configuration, validation, and error handling.
- Environment Configuration: API keys and application settings are securely managed using the `.env` file.

Workflow:

User
↓
▼

HTML/CSS Frontend



FastAPI Backend



Question Answering Module
Concept Explanation Module
Quiz Generation Module
Text Summarization Module
Learning Path Recommendation Module



GeminiAPI/AI Models



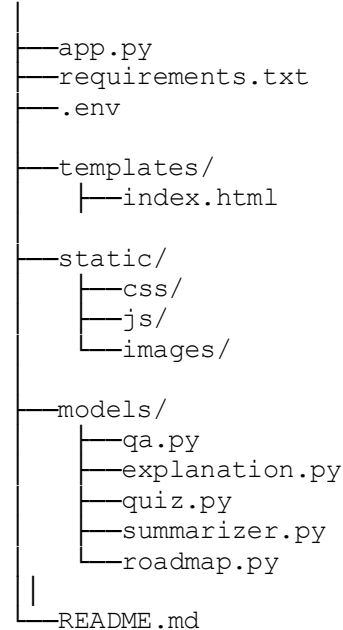
AI Response



Display Results to User

2. Project Structure

EduGenie/



3. Running the Application

Pre requisites

- Python3.10 or above
- FastAPI
- Uvicorn
- GeminiAPIKey
- Virtual Environment

Install Dependencies

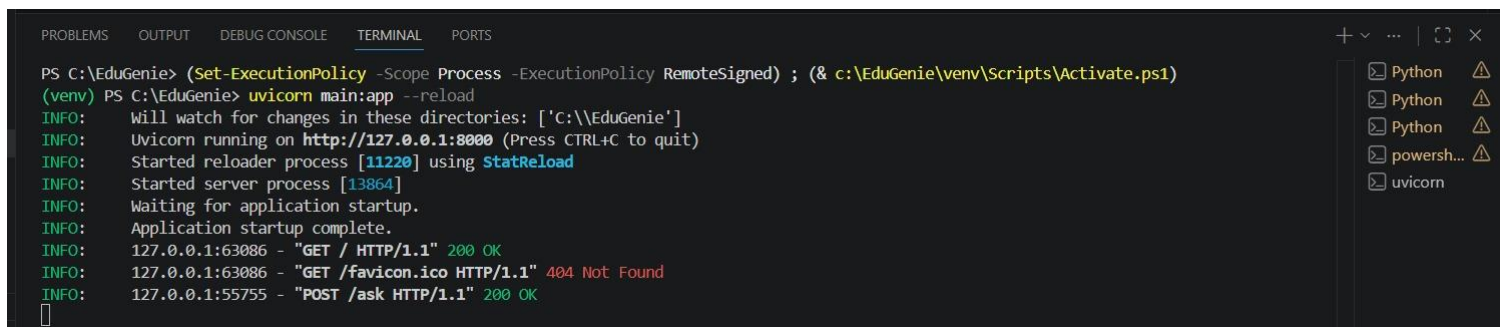
Pip install requirements.txt

Run the Application

```
uvicornapp:app--reload
```

Open the Application

<http://127.0.0.1:8000>

A screenshot of a Visual Studio Code terminal window. The terminal shows the command prompt for a PowerShell session in the directory C:\EduGenie. The user runs the command (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\EduGenie\venv\Scripts\Activate.ps1) to activate the virtual environment. Then, they run (venv) PS C:\EduGenie> uvicorn main:app --reload. The terminal output shows that Uvicorn is running on http://127.0.0.1:8000, it started a reloader process [11220] using StatReload, and a server process [13864]. It then shows several HTTP requests: a GET request for / HTTP/1.1 returning 200 OK, a GET request for /favicon.ico HTTP/1.1 returning 404 Not Found, and a POST request for /ask HTTP/1.1 returning 200 OK. On the right side of the terminal, there is a sidebar showing the file explorer with folders for Python, powershell, and uvicorn.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\EduGenie> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\EduGenie\venv\Scripts\Activate.ps1)
(venv) PS C:\EduGenie> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\EduGenie']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [11220] using StatReload
INFO: Started server process [13864]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:63086 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:63086 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 127.0.0.1:55755 - "POST /ask HTTP/1.1" 200 OK
```

The FastAPI server starts successfully, and the EduGenie home page is displayed in the web browser.

4. Application Running

The application launches successfully and provides a responsive interface where users can select different educational AI services. Navigation between modules is seamless, and responses are displayed in real time.

5. Application Work flow

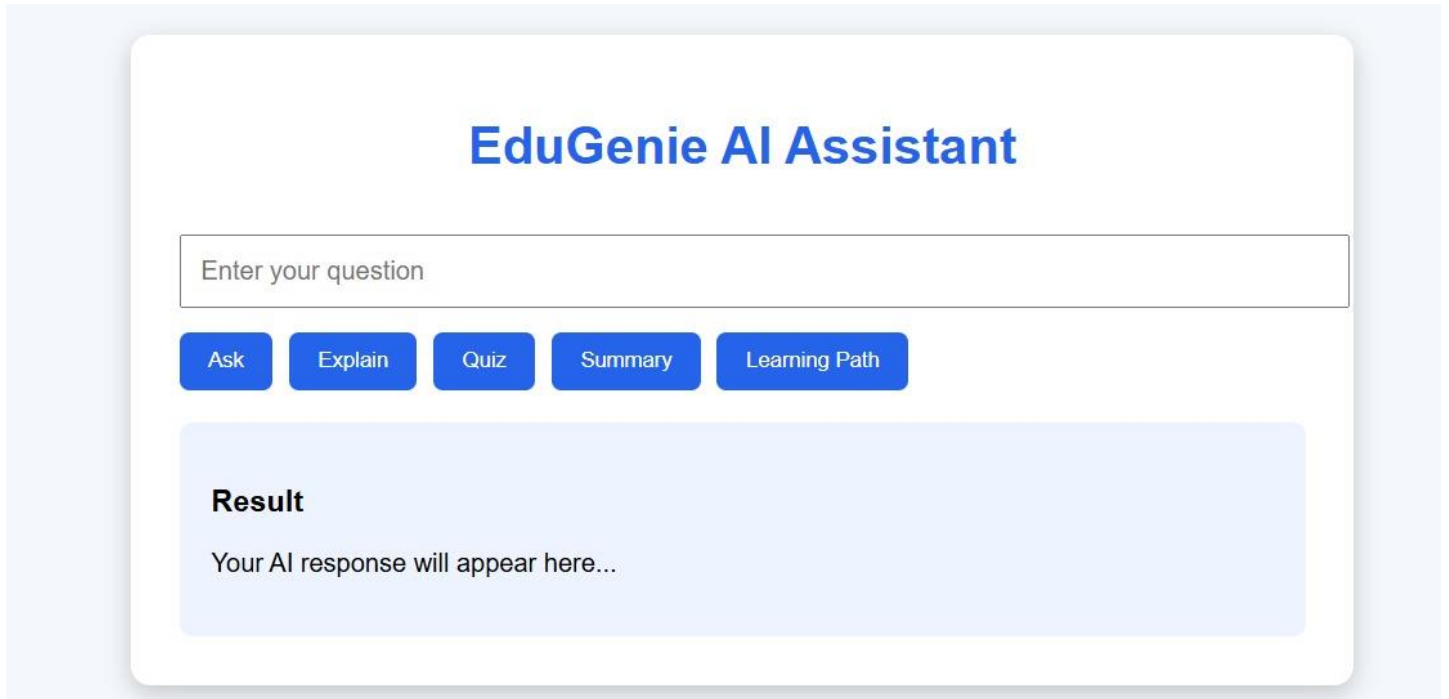


Figure1:Edu Genie Home Page

Displays them in interface with options for:

- Intelligent Question Answering
- Concept Explanation
- Quiz Generation
- Text Summarization
- Learning Path Recommendation

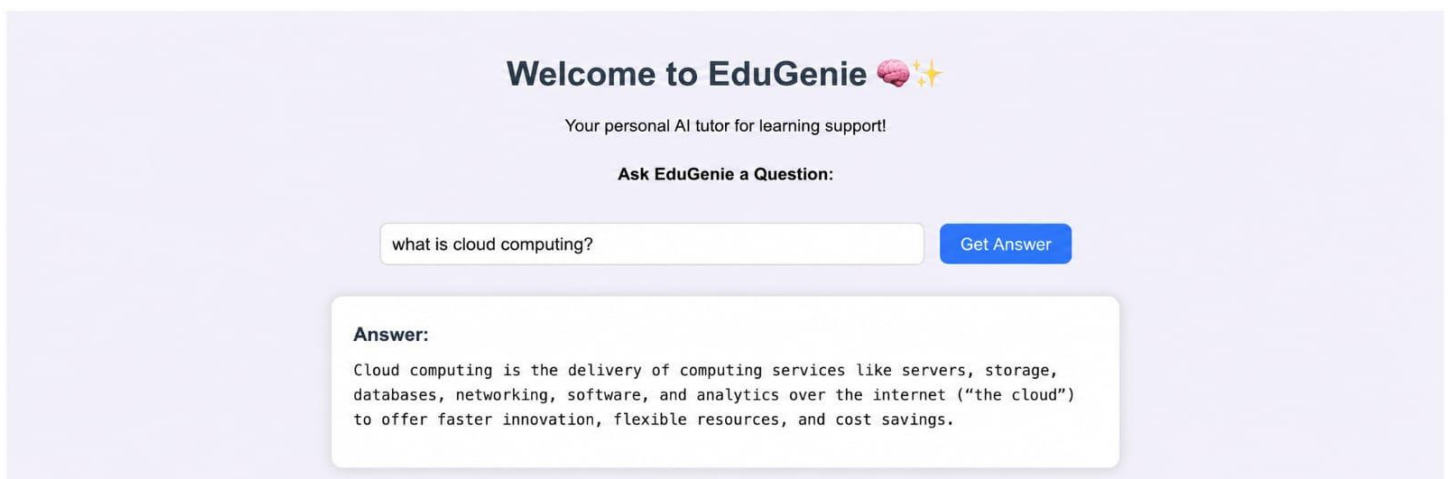


Figure2:Intelligent Question Answering

The user enters an educational question, and EduGenie generates an accurate answer with additional learning context.

Need an Explanation?

Pandas in Python

Explain

Explanation:

Pandas is a powerful Python library used for data manipulation and analysis. It provides data structures like Series (1D) and DataFrame (2D) to store and analyze data efficiently. With Pandas, you can easily load, clean, transform, analyze, and visualize data using simple functions. It is widely used in data science, machine learning, and data analysis projects.

Figure3:Concept Explanation

Users submit a topic or concept, and the system provides a simplified explanation suitable for learners.

Generate a Quiz:

Popular AI tools

Generate Quiz

Quiz:

Q1: Which of the following is a popular AI tool used for conversational AI and chatbots?

☐ TensorFlow

☐ Midjourney

☐ OpenAI ChatGPT

☐ Figma

Check Answer

Figure4:AI Quiz Generation

The application generates topic-specific multiple-choice questions for self-assessment.

Summarize a Paragraph:

Artificial Intelligence (AI) is transforming the world by enabling machines

Summarize

Summary:

Artificial Intelligence is revolutionizing the world by enabling machines to learn, reason, and make decisions. It is used in various fields like healthcare, education, transportation, and business to automate tasks, improve efficiency, and solve complex problems, making our lives easier and smarter.

Figure5:Educational Text Summarization

Long educational content is summarized into concise and easy-to-understand key points.

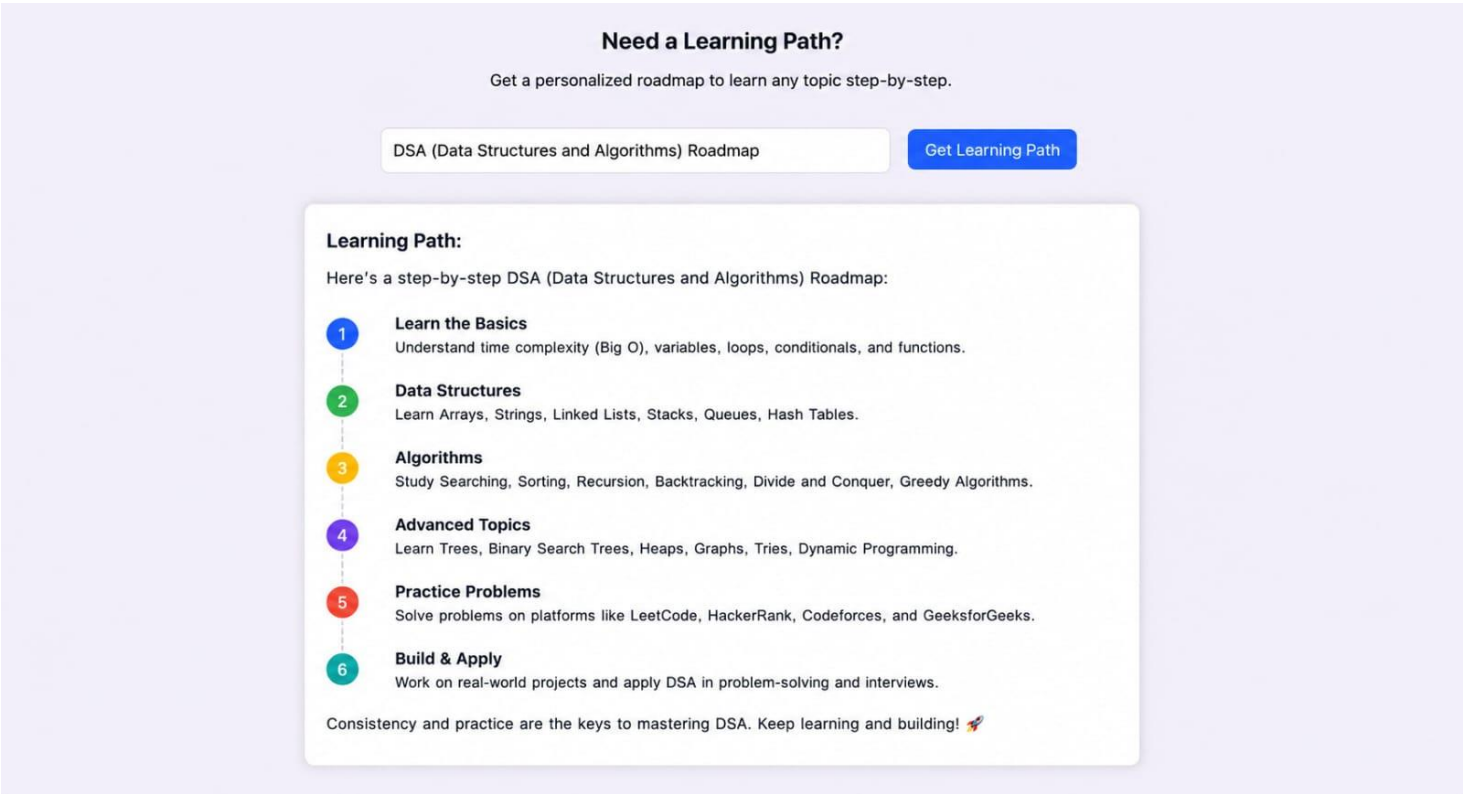


Figure6:Personalized Learning Path

The system creates a structured learning roadmap covering beginner, intermediate, and advanced topics with recommended study progression.